

Milestone 1: Data Preparation & Initial Analysis – Summary Document

1. Dataset Source:

- Global Weather Repository (Kaggle)
- Manually downloaded and extracted into the Jupyter Notebook environment.

2. Dataset Schema:

- Includes fields such as:
- country, location_name, latitude, longitude
- timezone, last_updated, temperature_celsius, temperature_fahrenheit
- condition_text, wind_mph, wind_kph, pressure_mb, precip_mm, humidity
- visibility_km, uv_index, air_quality indices, sunrise, sunset, etc.

3. Key Variables Used:

- Temperature (temperature_celsius)
- Humidity
- Wind Speed (wind_kph)
- Precipitation (precip_mm)
- Pressure (pressure_mb)
- Timestamp (last_updated)

4. Data Quality Issues Identified:

- Missing values in humidity and other numeric fields.
- Missing or invalid timestamps in last_updated.
- Potential anomalies in temperature, precipitation, and wind values.

5. Data Cleaning Steps:

- Removed rows with missing temperature_celsius.
- Filled missing humidity using median.
- Filled remaining numeric missing values using median.
- Converted last_updated to datetime and removed invalid entries.

- Removed negative precipitation and wind values.
- Filtered unrealistic temperature ranges ($< -80^{\circ}\text{C}$ or $> 60^{\circ}\text{C}$).

6. Unit Conversions:

- Fahrenheit retained but Celsius used as primary temperature unit.
- wind_kph used as the primary wind metric.

7. Data Aggregation:

- Extracted year and month from last_updated.
- Grouped by year and month to compute monthly averages.
- Saved cleaned dataset to: data/cleaned/monthly_weather.csv

8. Final Output:

- A clean, preprocessed, anomaly-free dataset ready for Milestone 2 analysis.